List of Experiments

Total No of Labs: 14

S. No.	Programs List	No. of Labs
CAD Experiments		
1	Study of basic graphics functions defined in “graphics.h”.	1
2	To implement Digital differential analyzer (DDA) Algorithm for drawing a line segment between two given end points A (x1, y1) and B(x2, y2).	1
3	To implement Bresenham’s line drawing algorithm for drawing a line segment between two given endpoints A (x1, y2) and B(x2, y2).	1
4	To implement Bresenham’s Cicrcle drawing algorithm for drawing a circle with a given center of circle P(x, y) and radius r.	1
5	To apply the basic 2D transformations such as translation, Scaling, Rotation, shearing and reflection for a given 2D object.	2
6	To apply the basic transformations such as translation, Scaling, Rotation for a given 3D object.	2
7	To generate a smooth curve by using Bezier curve technique for a given set of control points.	1
8	To implement Cohen–Sutherland 2D clipping and window–viewport mapping	1
Product Design		
9	Introduction to CAD software’s.	1
10	Generation of any one simple solid model using any CAD software.	1
11	Generation of any one assembly of mechanical part using any CAD software.	1
12	Design and develop the complete lifecycle for any product.	1

Experiment 1

Aim: Study of basic graphics functions defined in "graphics.h".

Theory:

What is computer graphics?

- ***Computer graphics* deals with all aspects of creating images with a computer**

  Hardware

  Software

  Applications

Computer graphics are graphics created using computers and, more generally, the representation and manipulation of image data by a computer. The development of computer graphics, or simply referred to as CG, has made computers easier to interact with, and better for understanding and interpreting manytypes of data. Developments in computer graphics have had a profound impact on many types of media and have revolutionized the animation and video game industry. Typically, the term *computer graphics* refers to several different things: · the representation and manipulation of image data by a computer · the various technologies used to create and manipulate images · the images so produced and The subfield of computer science which studies methods for digitally synthesizing and manipulating visual content **Applications:** · Computer Aided Design · Computer simulation · Digital art · Education · Graphic design · Information visualization · Scientific visualization · Video Games · Virtual reality · Web design

BASIC GRAPHICS FUNCTION

1) INITGRAPH

- Initializes the graphics system.

Declaration

- Void far initgraph(int far *graphdriver)

Remarks

- To start the graphic system, you must first call initgraph.
- Initgraph initializes the graphic system by loading a graphics driver from disk (or validating a registered driver) then putting the system into graphics mode.
- Initgraph also resets all graphics settings (color, palette, current position, viewport, etc) to their defaults then resets graph.

2) GETPIXEL, PUTPIXEL

- Getpixel gets the color of a specified pixel.
- Putpixel places a pixel at a specified point.

Decleration

- Unsigned far getpixel(int x, int y)
- Void far putpixel(int x, int y, int color)

Remarks

- Getpixel gets the color of the pixel located at (x,y);
- Putpixel plots a point in the color defined at (x, y).

Return value

- Getpixel returns the color of the given pixel.
- Putpixel does not return.

3) CLOSE GRAPH

- Shuts down the graphic system.

Decleration

- Void far closegraph(void);

Remarks

- Close graph deallocates all memory allocated by the graphic system.
- It then restores the screen to the mode it was in before you called initgraph.

Return value

- None.

4) ARC, CIRCLE, PIESLICE

- arc draws a circular arc.
- Circle draws a circle
- Pieslice draws and fills a circular pieslice

Decleration

- Void far arc(int x, int y, int stangle, int end_angle, int radius);
- Void far circle(int x, int y, int radius);
- Void far pieslice(int x, int y, int stangle, int end_angle, int radius);

Remarks

- Arc draws a circular arc in the current drawing color
- Circle draws a circle in the current drawing color
- Pieslice draws a pieslice in the current drawing color, then fills it using the current fill pattern and fill color.

5) ELLIPSE, FILLELIPSE, SECTOR

- Ellipse draws an elliptical arc.
- Fillellipse draws and fills ellipse.
- Sector draws and fills an elliptical pie slice.

Decleration

- Void far ellipse(int x, int y, int stangle, int end_angle, int xradius, int yradius)
- Void far fillellipse(int x, int y, int xradius, int yradius)
- Void farsectoe(int x, int y, int stangle, int end_angle, int xradius, int yradius)

Remarks

- Ellipse draws an elliptical arc in the current drawing color.

- Fillellipse draws an elliptical arc in the current drawing color and than fills it with fill color and fill pattern.
- Sector draws an elliptical pie slice in the current drawing color and than fills it using the pattern and color defined by setfillstyle or setfillpattern.

6) FLOODFILL

- Flood-fills a bounded region.

Decleration

- Void far floodfill(int x, int y, int border)

Remarks

- Floodfills an enclosed area on bitmap device.
- The area bounded by the color border is flooded with the current fill pattern and fill color.
- (x,y) is a "seed point"
    - If the seed is within an enclosed area, the inside will be filled.
    - If the seed is outside the enclosed area, the exterior will be filled.
- Use fillpoly instead of floodfill wherever possible so you can maintain code compatibility with future versions.
- Floodfill doesnot work with the IBM-8514 driver.

Return value

- If an error occurs while flooding a region, graph result returns '1'.

7) GETCOLOR, SETCOLOR

- Getcolor returns the current drawing color.
- Setcolor returns the current drawing color.

Decleration

- Int far getcolor(void);
- Void far setcolor(int color)

Remarks

- Getcolor returns the current drawing color.
- Setcolor sets the current drawing color to color, which can range from 0 to getmaxcolor.
- To set a drawing color with setcolor , you can pass either the color number or the equivalent color name.

8) LINE, LINEREL, LINETO

- Line draws a line between two specified pints.
- Onerel draws a line relative distance from current position(CP).
- Linrto draws a line from the current position (CP) to(x,y).

Decleration

- Void far lineto(int x, int y)

Remarks

- Line draws a line from (x1, y1) to (x2, y2) using the current color, line style and thickness. It does not update the current position (CP).
- Linerel draws a line from the CP to a point that is relative distance (dx, dy) from the CP, then advances the CP by (dx, dy).
- Lineto draws a line from the CP to (x, y), then moves the CP to (x,y).

Return value

- None

9) RECTANGLE

- Draws a rectangle in graphics mode.

Decleration

- Void far rectangle(int left, int top, int right, int bottom)

Remarks

- It draws a rectangle in the current line style, thickness and drawing color.
- (left, top) is the upper left corner of the rectangle, and (right, bottom) is its lower right corner.

Return value

- None.

Note: Algorithms of some simple programs have also been included.

Here is a sample program that initializes the graphics mode in C Language.

```
#include<graphics.h>
#include<conio.h>
main()
{
int gd=DETECT,gm;
initgraph(&gd,&gm,""); /* initialization of graphic mode */
circle(150,150,100);
getch();
 closegraph(); /* Restore orignal screen mode */
} /* End of program */
```